

An Investigation of Bayesian Lasso Priors for Linear Mean Gaussian Process Regression

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1 Introduction

Gaussian process models are stochastic models that define a probability distribution over functions. Because of their ability to model uncertainty and capture complex, nonlinear relationships, Gaussian processes are valuable non-parametric tools for both regression and classification tasks.

In Gaussian process regression, we begin with a set of n observations in the form $\mathcal{D} = \{X, \mathbf{y}\}$, where $\mathbf{y}$ is an $n \times 1$ vector of scalar responses, and X is an $n \times p$ matrix of input features. Each individual observation can be represented by $(\mathbf{x}_i, y_i)$. The Gaussian process model is represented by

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$$

where the mean function is $m(\mathbf{x}) = E[f(\mathbf{x})]$ and the covariance function is $k(\mathbf{x}, \mathbf{x}') = E[(f(\mathbf{x}) - m(\mathbf{x}))(f(\mathbf{x}') - m(\mathbf{x}'))]$. The covariance function, or kernel, is used to encode assumptions about the similarity between data points based on their features (Rasmussen and Williams 2006).

Frequently, Gaussian process means are set to zero for simplicity (Rasmussen and Williams 2006). However, a linear mean function can be chosen to improve interpretability, as a Gaussian process with a linear mean function allows for both insight into the model through the linear coefficients and additional flexibility to capture nonlinear structure within the residuals. With a linear mean function, regularization can be used to encourage sparsity in the linear coefficients, through which information about feature relevance can be gained.

However, incorporating regularization of the linear mean into the fitting of a Gaussian process is not common. Regularization of Gaussian processes is often handled through the kernel function or through restrictive priors on the model hyperparameters (Rasmussen and Williams 2006). While this can be successful in reducing overfitting, it does not lead to interpretable results regarding feature relevance.

Park and Casella (2008) provide a framework for a Bayesian analogy to an ℓ_1 -penalized least squares estimate for linear regression coefficients using Laplace priors on regression coefficients. The Bayesian Lasso establishes a method of incorporating ℓ_1 -like regularization through prior distributions placed on the linear coefficients. In this study, this structure was adapted for the linear mean function of a Gaussian process regression model. Using Bayesian inference to draw samples and estimate posterior means for the linear model coefficients and Gaussian process hyperparameters also allows the inclusion of known prior distributions and a quantification of the uncertainty within the point estimates.

However, a significant challenge of this structure is that the Gaussian process component and the linear function component can compete to explain the same patterns in the data. As the flexibility of the Gaussian process allows it to capture variation that the linear mean might be able to explain, this obscures the interpretability of the model. This creates a limitation for feature selection with the Bayesian Lasso technique. Even if the Bayesian Lasso shrinks some coefficients, those resulting coefficients may not be meaningful if the Gaussian process is capable of explaining the same signal.

Plumlee and Roshan (2016) introduce an orthogonal Gaussian process approach to address the interpretability problem. They define a modified kernel $k_*(\mathbf{x}, \mathbf{x}')$ formed by subtracting the Gaussian process covariance component that is aligned with the linear mean $m(x)$ from the original kernel function $k(\mathbf{x}, \mathbf{x}')$. This kernel $k_*(\mathbf{x}, \mathbf{x}')$ then restricts the Gaussian process component to be orthogonal to the mean function, which reduces the overlap between the two components.

This study investigates whether Bayesian Lasso priors, alone or in combination with orthogonal Gaussian process methods, can recover interpretability from a linear mean function within Gaussian process regression models.

2 Methodology

2.1 Problem Set Up

In Gaussian process regression with a linear mean, the data is modeled using the following equation

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + g(\mathbf{x}_i) + \eta$$

In this equation, $\mathbf{x}_i^\top \boldsymbol{\beta}$ represents the linear mean component, $g(x) \sim \mathcal{GP}(m(\mathbf{x}) = 0, k(\mathbf{x}, \mathbf{x}'))$ represents a non-linear function modeled by a zero-mean Gaussian process, and $\eta \sim \mathcal{N}(0, \sigma_n^2)$ represents independent Gaussian noise.

To fit a Gaussian process to observed data, the loss function can be optimized or evaluated during Bayesian sampling. The loss function is given by the log marginal likelihood of a multivariate normal distribution with linear mean $\mu = X\boldsymbol{\beta}$.

$$\ell(\theta, \sigma_n^2, \boldsymbol{\beta}; X, y) = -\frac{n}{2} \log(2\pi) - \frac{1}{2} \log |C| - \frac{1}{2} (y - X\boldsymbol{\beta})^T C^{-1} (y - X\boldsymbol{\beta})$$

To reflect assumptions that observations will contain random Gaussian noise, we use $C = K + \sigma_n^2 I_n$ as the total covariance matrix, composed of the kernel matrix K and the noise variance σ_n^2 multiplied by the identity matrix (Rasmussen and Williams 2006).

In this study, the Squared Exponential or Radial Basis Function kernel is used, which takes the following form.

$$K_{ij} = \sigma_{GP}^2 \exp \left(-\frac{1}{2} \left\| \frac{x_i - x_j}{\ell} \right\|^2 \right)$$

This is the most commonly-used kernel and has the key feature of being infinitely differentiable, which means that it produces smooth functions. The smoothness is controlled by the lengthscale hyperparameter ℓ , where a small value reflects more localized correlation between observations, producing a more wiggly curve, and a large value reflects wider correlation between observations, producing a smoother curve. The signal variance hyperparameter σ_{GP}^2 scales the kernel and reflects the variation between the function values and the mean, with larger values allowing for greater vertical spread of the function values (Rasmussen and Williams 2006).

To allow for different features to influence the underlying process at different scales, each feature is allowed to have its own lengthscale in the kernel. This is modeled through a kernel with separate lengthscales per active feature, referred to as an Automatic Relevance Determination (ARD) kernel. This takes the following form, where $\boldsymbol{\ell}$ corresponds to a vector of individual lengthscales (Rasmussen and Williams 2006).

$$K_{ij} = \sigma_{GP}^2 \exp \left(-\frac{1}{2} \left\| \frac{x_i - x_j}{\boldsymbol{\ell}} \right\|^2 \right)$$

2.2 Data

Models were tested on a dataset from a diabetes study from Efron et al. (2004). This dataset has 442 observations, corresponding to individual patients. Each patient had 10 baseline measures recorded: age, sex, body mass index, blood pressure, and six blood serum measurements. The response variable corresponds

to a measure of diabetes progression one year after the initial measurements were observed. Features are centered around the mean and scaled by the standard deviation multiplied by the square root of $n = 442$ (Efron et al. 2004). Pre-standardized data was loaded directly via a scikit-learn Python package (Pedregosa et al. 2015). Following the structure in Park and Casella (2008), the response variable was centered about the mean so that the intercept could be marginalized, which simplifies computation. Models were trained on a random split, with 80% of the data used for training and 20% reserved to test each model’s ability to generalize to unseen data.

2.3 Models

2.3.1 Bayesian Lasso Baseline

The first model analyzed in this study is the Bayesian Lasso from Park and Casella (2008). It incorporates an ℓ_1 -like penalty on a linear function without a Gaussian process component. This model stands as a baseline against which the Bayesian regularization approaches can be evaluated.

The Bayesian Lasso places independent and identical Laplace priors on regression coefficients, conditioned on the variance σ_n^2 , in the form shown below.

$$\pi(\boldsymbol{\beta} \mid \sigma_n^2) = \prod_{j=1}^p \frac{\lambda}{2\sqrt{\sigma_n^2}} e^{-\lambda|\beta_j|/\sqrt{\sigma_n^2}}$$

Through this, the hierarchical model that places an ℓ_1 -analogous penalty on the linear coefficients $\boldsymbol{\beta}$ of the mean function can be defined as follows.

$$y \mid \mu, X, \boldsymbol{\beta}, \sigma_n^2 \sim N_n(\mu \mathbf{1}_n + X\boldsymbol{\beta}, \sigma_n^2 I_n),$$

$$\boldsymbol{\beta} \mid \sigma_n^2, \tau_1^2, \dots, \tau_p^2 \sim N_p(\mathbf{0}_p, \sigma_n^2 D_\tau)$$

$$D_\tau = \text{diag}(\tau_1^2, \dots, \tau_p^2)$$

$$\sigma_n^2, \tau_1^2, \dots, \tau_p^2 \sim \pi(\sigma_n^2) d\sigma_n^2 \prod_{j=1}^p \frac{\lambda^2}{2} e^{-\lambda^2 \tau_j^2 / 2} d\tau_j^2$$

$$\sigma^2, \tau_1^2, \dots, \tau_p^2 > 0$$

Thus, this model contains the parameters given in the table below, with the corresponding prior distributions assigned. Note that values α_{λ^2} and β_{λ^2} come from Park and Casella (2008)), and that the overall mean μ is marginalized though centering the response variable.

Regularization parameter	$\lambda^2 \sim \text{Gamma}(\alpha_{\lambda^2} = 1.0, \beta_{\lambda^2} = 1.78)$
Local shrinkage parameters	$\tau_1^2, \dots, \tau_p^2 \sim \text{Exp}\left(\frac{\lambda^2}{2}\right)$
Linear mean coefficients	$\beta \sim \mathcal{N}\left(\mu_\beta = 0, \sigma_\beta = \sqrt{(\sigma_n^2 \tau^2)}\right)$
Observation noise	$\sigma_n \sim \text{Half Normal}(\sigma = 1.0)$

Table 1: Parameters and prior distributions for Bayesian Lasso

2.3.2 Gaussian Process with Bayesian Lasso

Applying the Bayesian Lasso framework to the linear mean function of a Gaussian process allows for ℓ_1 -like regularization of the linear mean function. The sampler uses standard Gaussian process likelihood with a linear mean. Here, the penalty term λ is not added to the likelihood, because the priors placed on the linear mean serve the same purpose. λ controls the amount of regularization through the hierarchical model. This parameter λ defines the distribution for $\tau_1^2, \dots, \tau_p^2$, which in turn influence the distribution for linear coefficients β .

The regularization hyperparameter can be fixed or sampled through a given prior distribution. When sampled, a Gamma distribution is used, following the same form given by Park and Casella (2008). Note that all other prior distributions match that of the Bayesian Lasso method.

This model contains the following parameters, with the corresponding prior distributions assigned.

Global shrinkage parameters	$\ell \sim \log \mathcal{N}(\mu_\ell = -2, \sigma_\ell = 1)$
Regularization parameter	$\lambda^2 \sim \text{Gamma}(\alpha_{\lambda^2} = 1.0, \beta_{\lambda^2} = 1.78)$
Local shrinkage parameters	$\tau_1^2, \dots, \tau_p^2 \sim \text{Exp}\left(\frac{\lambda^2}{2}\right)$
Linear mean coefficients	$\beta \sim \mathcal{N}\left(\mu_\beta = 0, \sigma_\beta = \sqrt{(\sigma_n^2 \tau^2)}\right)$
Gaussian process standard deviation	$\sigma_{GP} \sim \text{Inv Gamma}(\alpha = 3.0, \beta = 1.0)$
Observation noise	$\sigma_n \sim \text{Half Normal}(\sigma = 1.0)$

Table 2: Parameters and prior distributions for Gaussian process with Bayesian Lasso

2.3.3 Orthogonal Gaussian Process with Analytical β

Plumlee and Roshan (2016) introduce a modified covariance function k_* that orthogonalizes the Gaussian process component with respect to the linear mean function.

Here, orthogonality means that the Gaussian process is prevented from explaining variation that is in the span of the linear mean basis $g(\mathbf{x})$. Since the mean function is a linear function with no intercept, we use $g(\mathbf{x}) = \mathbf{x}$. This means that the Gaussian process is restricted from variation that can be represented by

linear combinations of columns of X . The modified covariance function $k_*(\mathbf{x}, \mathbf{x}')$ is formed by subtracting the component of the Gaussian process covariance that can be represented by these linear combinations from the original kernel function $k(\mathbf{x}, \mathbf{x}')$.

Plumlee and Roshan (2016) define

$$h(x) = \int_{\mathcal{X}} k(x, \xi) g(\xi) d\xi$$

and

$$H = \int_{\mathcal{X}} \int_{\mathcal{X}} k(\xi', \xi) g(\xi) g(\xi')^\top d\xi d\xi'$$

Because $g(\mathbf{x}) = \mathbf{x}$, these become

$$\int_{\mathcal{X}} k(x, \xi) \xi d\xi$$

and

$$H = \int_{\mathcal{X}} \int_{\mathcal{X}} k(\xi', \xi) \xi \xi'^\top d\xi d\xi'$$

Given that $k(\mathbf{x}, \mathbf{x}')$ is a covariance function that is bounded and continuous on $\mathcal{X} \times \mathcal{X}$ and H is a positive definite matrix, the modified covariance function is

$$k_*(\mathbf{x}, \mathbf{x}') = k(\mathbf{x}, \mathbf{x}') - h(\mathbf{x})^T H^{-1} h(\mathbf{x}')$$

This produces a covariance matrix K_* . To reflect assumptions of observation noise, our final model uses $C_* = K_* + \sigma_n I_n$.

Given C_* , estimates of linear coefficients $\hat{\beta}$ can be constructed analytically using

$$\hat{\beta} = (G^T C_*^{-1} G)^{-1} G^T C_*^{-1} \mathbf{y}$$

Because $g(\mathbf{x}) = \mathbf{x}$, $G = X$, so we use

$$\hat{\beta} = (X^T C_*^{-1} X)^{-1} X^T C_*^{-1} \mathbf{y}$$

This approach is designed with the goal of recovering the interpretability of the linear mean function by preventing competition between the Gaussian process and linear components. In this analytical version, the linear coefficients are estimated from sampled hyperparameters, rather than sampled directly from a posterior distribution. Therefore, this version does not yet apply Bayesian Lasso priors over the orthogonal

Gaussian process mean function.

This model contains the following parameters, with the corresponding prior distributions assigned.

Global shrinkage parameters	$\ell \sim \log \mathcal{N}(\mu_\ell = -2, \sigma_\ell = 1)$
Regularization parameter	$\lambda^2 \sim \text{Gamma}(\alpha_{\lambda^2} = 1.0, \beta_{\lambda^2} = 1.78)$
Gaussian process standard deviation	$\sigma_{GP} \sim \text{Inv Gamma}(\alpha = 3.0, \beta = 1.0)$
Observation noise	$\sigma_n \sim \text{Half Normal}(\sigma = 1.0)$

Table 3: Parameters and prior distributions for orthogonal Gaussian process with analytical β

2.3.4 Orthogonal Gaussian Process with Bayesian Lasso

The final model implemented in this study is an orthogonal Gaussian process that incorporates the Bayesian Lasso framework as it samples the linear coefficients β . This model uses the kernel $k_*(\mathbf{x}, \mathbf{x}')$ introduced previously, as well as the hierarchical model of the Bayesian Lasso to place an ℓ_1 -analogous penalty on the linear mean coefficients.

This model contains the following parameters, with the corresponding prior distributions assigned.

Global shrinkage parameters	$\ell \sim \log \mathcal{N}(\mu_\ell = -2, \sigma_\ell = 1)$
Regularization parameter	$\lambda^2 \sim \text{Gamma}(\alpha_{\lambda^2} = 1.0, \beta_{\lambda^2} = 1.78)$
Linear mean coefficients	$\beta \sim \mathcal{N}(\mu_\beta = 0, \sigma_\beta = \sqrt{(\sigma_n^2 \tau^2)})$
Gaussian process standard deviation	$\sigma_{GP} \sim \text{Inv Gamma}(\alpha = 3.0, \beta = 1.0)$
Observation noise	$\sigma_n \sim \text{Half Normal}(\sigma = 1.0)$

Table 4: Parameters and prior distributions for orthogonal Gaussian process with analytical β

2.4 Implementation

This study uses Bayesian inference with PyMC (Abril-Pla et al. 2023) to draw samples and estimate posterior means for both the linear model coefficients and Gaussian process hyperparameters. This method allows both the inclusion of known prior distributions and a quantification of the uncertainty within the point estimates, while avoiding optimization of a non-differentiable function.

Sampling is implemented locally in Python. Previous attempts used the Bilby package in Python, first run locally and then on the Quest High-Performance Computing Cluster at Northwestern University (Ashton et al. 2019). Due to the extensive time required for each run, and the difficulty of implementing custom Gaussian process likelihoods, the experiments were rewritten to use PyMC. This allowed the hierarchical structure to be preserved while sampling. Samples used the Metropolis-Hastings step method.

2.5 Evaluation Metrics

To evaluate results of each model, this study uses an assortment of both quantitative and visual methods to assess convergence and model predictions. To assess convergence, we use effective sample size (Bulk-ESS, Tail-ESS), potential scale reduction factor ($\hat{R}$), and trace plots. To assess predictive accuracy, we use root mean squared error on the training and testing datasets.

Effective sample size, or ESS, is defined as the number of effective simulation draws after accounting for autocorrelation within each chain. In practice, a minimum ESS of 10 per chain is recommended for the simulation to be able to converge to the target distribution, although some situations require more samples (Gelman et al. 2014). PyMC generates values for Bulk-ESS and Tail-ESS, which are more modern variants of ESS. Bulk-ESS assesses convergence for the bulk of the distribution, while tail-ESS assesses convergence of the 5% and 95% quantiles of the distribution. A threshold of 400 for both bulk-ESS and tail-ESS is recommended to assess if estimates are likely to closely approximate the target distributions (Vehtari et al. 2020).

Potential scale reduction factor, or $\hat{R}$, is an estimate of the scale of the current distribution for the sampled parameter compared to the distribution if the number of draws increased to infinity. It is used as a diagnostic to assess approximate convergence, if $\hat{R} \approx 1$, and to identify issues with convergence if $\hat{R} \not\approx 1.0$ (Gelman et al. 2014). PyMC generates values for $\hat{R}$ for each parameter after sampling.

Trace plots display the movement of the parameter estimates across samples. In a trace plot, mixing, stationarity, and no correlation between samples are the key features that show whether a simulation has converged. Good mixing means that chains overlap and are not separated for long periods of time. Stationarity means that the samples vary around a stable mean without any trends over time. Lack of correlation means that draws should be visibly independent of each other. A trace plot may not show stationary behavior right away, in which case the sampler might require a warm-up tuning period removed so that estimates are constructed using only the stabilized chains (Gelman et al. 2014).

This study uses root mean squared error (RMSE) to assess a model’s predictive accuracy.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE is computed on both the training and testing datasets. Low RMSE on training data suggests that the model captures the patterns within the training set effectively. Low RMSE on testing data reflects strong generalization to unseen data.

3 Results

3.1 Bayesian Lasso

The goal of the Bayesian Lasso linear model in this study is to establish a baseline for comparison on the standardized diabetes dataset with the subsequent Gaussian Process regression methods, following the structure presented in Park and Casella (2008) on the same dataset.

The linear model is first run for 10,000 samples with no tuning steps to inspect convergence. See Figure 1 below for an example of a trace plot for a linear coefficient within this first untuned run. In this plot, the six chains are initialized close to zero, quickly increase, and stabilize at a lower value after approximately 2,000 steps. At 2,000 steps, the chains show good mixing. Without zero tuning steps removed, $\hat{R} > 1.0$, Bulk-ESS is less than 400, and Tail-ESS is less than 400 for all sampled coefficients. This indicates that the model requires has not reached the target distribution, and so we apply a tuning period of approximately 2,000 steps so that posterior estimates are calculated from samples taken after the chains have stabilized.

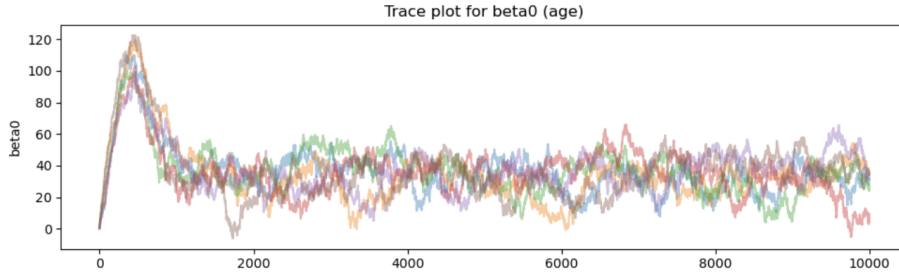


Figure 1: Trace for 'age' (β_0)

Next, the model is run with 2,000 tuning steps to capture posterior estimates of the parameters. Trace plots, such as the one shown in Figure 2 below, now show good mixing, a stable mean, and no visible patterns. For six of the ten linear coefficients, $\hat{R} = 1.0$, and both Bulk-ESS and Tail-ESS are greater than 400, indicating that the tuning phase significantly improved convergence. In contrast, parameters that still demonstrate higher $\hat{R}$ values also have significantly lower Bulk-ESS and Tail-ESS. Trace plots for these coefficients show that the issue may be related to correlation between consecutive samples. This implies that additional tuning steps are less likely to significantly improve their convergence.

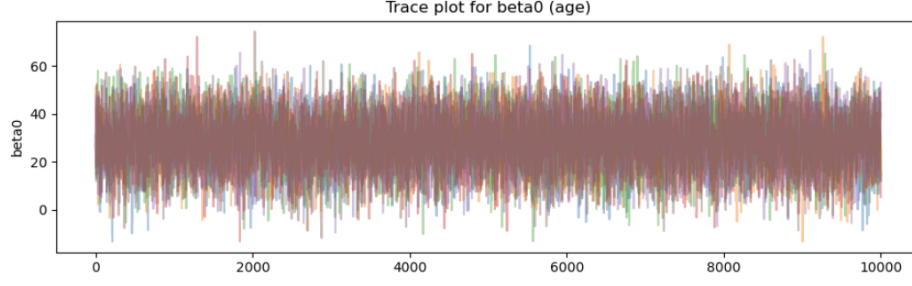


Figure 2: Trace for 'age' (β_0) with 2,000 tuning steps

The posterior estimates for the linear coefficients can be constructed from the samples, shown in Figure 3. 95% credible intervals for each estimate are constructed from posterior distributions, allowing us to quantify the uncertainty in each. In this model, only the credible interval for β_7 ('s3') crosses zero, meaning that the model is confident in the direction of the associations between the other nine features and the response variable. Intervals for many features are narrow, suggesting that the model is also confident about the relative magnitude of these associations.

From these linear coefficients, the model can be easily interpreted. For example, the model indicates that an increase in 's5' (β_9), at the time of measurement is associated with a significantly larger response value, meaning more severe disease progression after one year. Meanwhile, an increase in 's1' (β_5) is associated with a smaller response value, meaning less severe disease progress after one year.

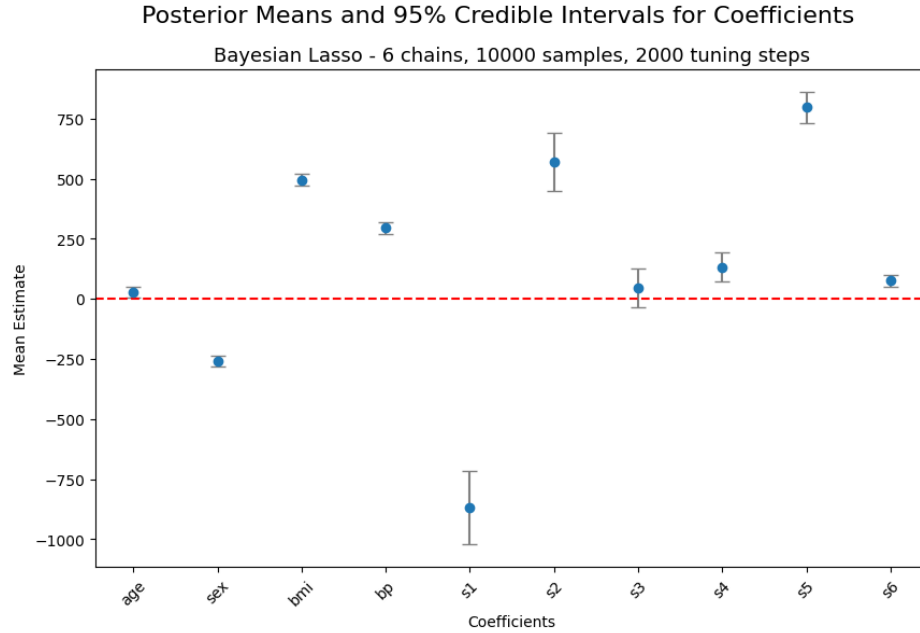


Figure 3: Posterior Means and 95% Credible Intervals for Linear Coefficients

In this experiment, the regularization parameter λ^2 is estimated to be 0.001 (95% Credible Interval: [0.000, 0.002]), which means that $\lambda = \sqrt{\lambda^2}$ is estimated to be 0.03 ([0.000, 0.045]). Because λ is close to zero, this demonstrates relatively weak shrinkage, meaning that the Bayesian Lasso is not strongly forcing the regression coefficients to zero.

To assess predictive performance, the posterior distribution of the RMSE values can be obtained by making predictions on the training and testing set from each sample. Using 10,000 samples and 2,000 tuning steps, we achieve a posterior mean estimate of 52.87 ([52.85, 52.89]) for the training set, and 56.40 ([56.12, 56.70]) for the testing set. The 95% credible intervals represent uncertainty in the estimate, where larger intervals correspond to high uncertainty.

These results establish a baseline for several factors, including posterior mean estimates and credible intervals for the linear coefficients, strength of the regularization parameter, prediction error, and interpretability. This enables comparison with further models on this data set.

3.2 Gaussian Process with Bayesian Lasso Priors

The next model implements Bayesian Lasso priors within the linear mean function of a Gaussian process, applying ℓ_1 -like regularization and testing whether interpretability is gained.

As in the previous model, the Gaussian process with Bayesian Lasso priors is run on the standardized diabetes data for 10,000 samples with no tuning period to learn how estimates move across samples. However, where the Bayesian Lasso linear model showed stabilization after 2,000 steps, this model shows relatively stable behavior across all samples for each parameter.

Values for $\hat{R}$ fall within [1.02, 1.05] for the ten linear coefficients. Bulk-ESS is above 400 for all but two coefficients, but Tail-ESS is lower, with many coefficients below 400. These diagnostics indicate that there are some issues with convergence in this model. As shown in Figure 4, the linear coefficients are generally concentrated around zero, but at irregular intervals, one chain will make a sudden large excursion and then later rejoin the other chains. This indicates that the chains occasionally get stuck exploring a region of the search space before returning to the group. From a visual inspection of the trace plot, we see that a tuning period would not meaningfully improve convergence, because the chain is stable from the start, even though individual chains show occasional spikes.

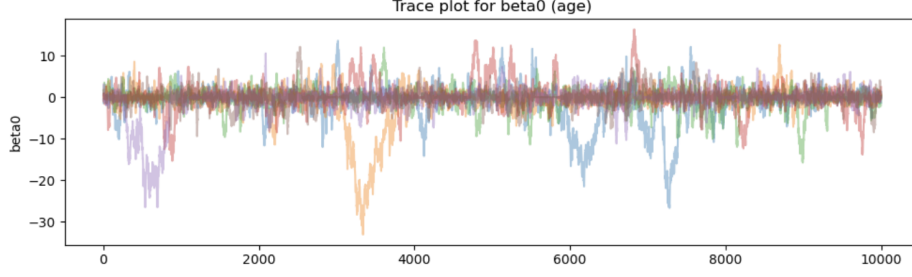


Figure 4: Trace for 'age' (β_0)

In this experiment, the regularization parameter λ^2 is estimated to be 0.528 ([0.001, 1.653]), with $\lambda = \sqrt{\lambda^2}$ estimated to be 0.727 ([0.032, 1.286]). In comparison with the estimate of $\lambda = 0.03$ from the linear model, this demonstrates a significant increase in regularization strength. This means that the Bayesian Lasso priors are forcing coefficients to zero. In Figure 5, which shows posterior mean estimates and 95% credible intervals for the linear coefficients, it is visible that the estimate for each coefficient is very close to zero, with credible intervals spanning wide regions on either side of zero.

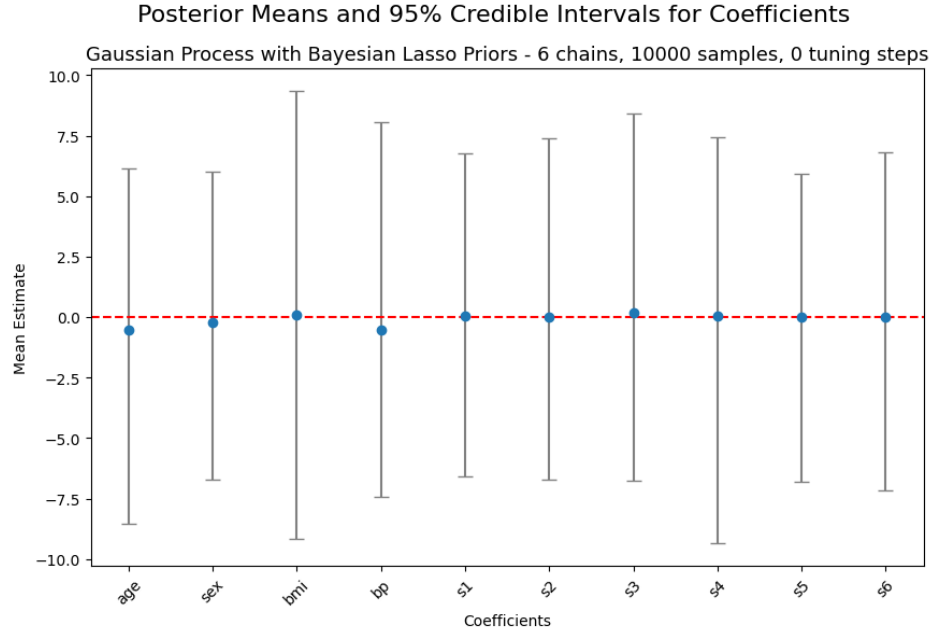


Figure 5: Posterior Means and 95% Credible Intervals for Linear Coefficients

The posterior distributions of the RMSE for the training and testing set are calculated from each draw. With 10,000 samples and no tuning steps, the posterior mean estimate of RMSE on the training set is 47.10 ([19.38, 60.86]) and 62.59 ([55.19, 73.12]) on the testing set. While the training error is lower than the Bayesian Lasso linear model, the estimate for testing error is slightly higher. For both sets, the uncertainty

of the Gaussian process with Bayesian Lasso priors is larger than the linear model.

The combination of a larger regularization penalty estimate and smaller error on the training set indicates that the model is pushing linear coefficients to zero and relying on the Gaussian process component of the model to explain all of the variation in the data.

Because the model forces all coefficients close to zero, interpretability of the linear mean is lost. Both the Gaussian process and the linear mean are capable of explaining similar signals in the data, and this model uses the Gaussian process component to explain all of it. Figure 6 illustrates this by separately plotting the linear mean and Gaussian process components of the regression model against the ground truth. For both the training and testing tasks, the linear component of the model contributes insignificantly to the predictions, which are the sum of the corresponding linear and Gaussian process components.

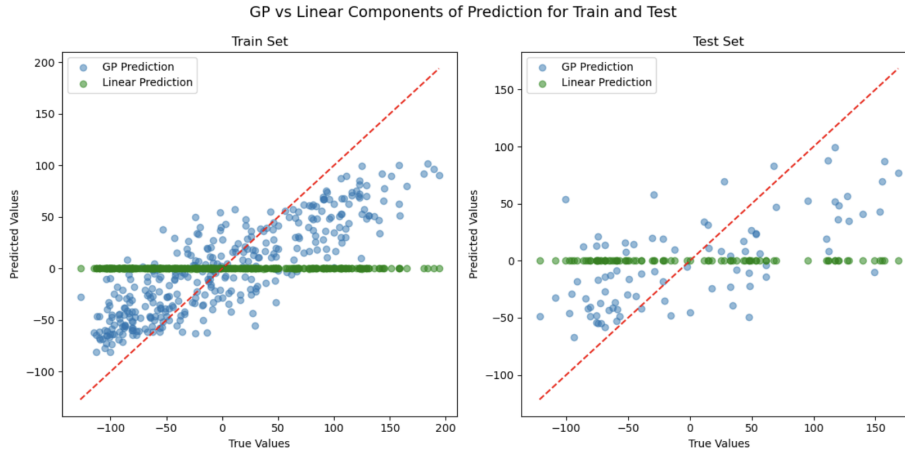


Figure 6: Gaussian Process and Linear Components of the Regression Task for Train and Test

This suggests that competition between the Gaussian process and the linear mean function to explain the same variation in the data, which leads to over-regularization of the linear coefficients, may present a major obstacle to recapturing interpretability through Bayesian Lasso priors within Gaussian processes.

Fixed Penalty Strength By fixing the regularization hyperparameter λ at the value of its posterior mean estimate from the linear model ($\lambda = 0.03$), we can assess the performance of the Gaussian process with Bayesian Lasso priors given weaker regularization.

With 10,000 steps and no tuning, we observe that trace plots, such as the example shown in Figure 7, show that weaker regularization leads to increased wandering of individual chains within the search space, as well as a wider search space for all chains. Chains all take separate paths, demonstrating poor mixing. Although the coefficients are no longer heavily-regularized, chains do not show clear convergence, although

they do remain approximately centered around zero. For the sampled linear coefficients, $\hat{R} > 1.2$, with both Bulk-ESS and Tail-ESS only showing values less than 100. These metrics show very poor convergence.

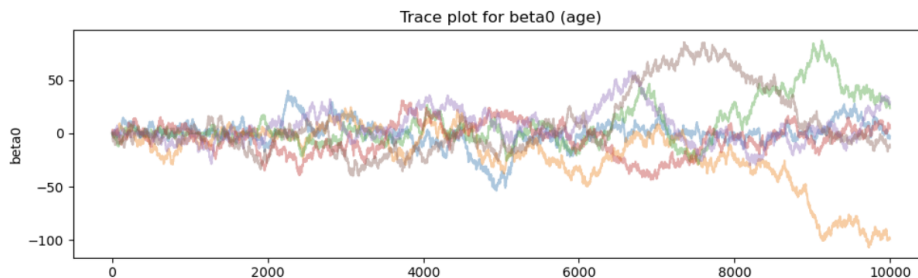


Figure 7: Trace for 'age' (β_0), given $\lambda = 0.03$

Posterior estimates for β move slightly away from zero, due to the wandering resulting from less regularization, and credible intervals become wider by a factor of nearly 10. All intervals include zero, shown in Figure 8, meaning that this model can make no conclusions about the direction or even relative magnitude of the associations between the features and the response.

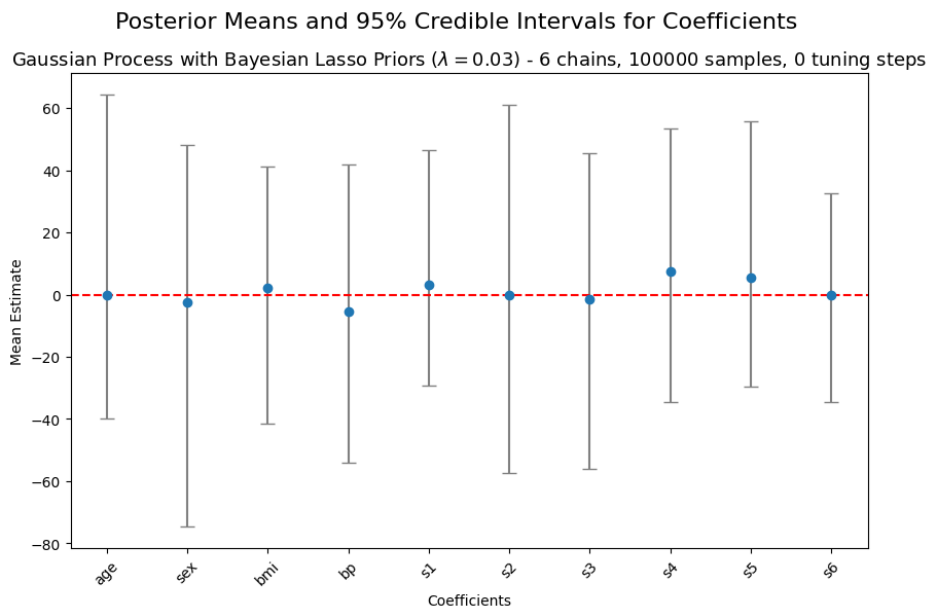


Figure 8: Posterior Means and 95% Credible Intervals for Linear Coefficients

With the regularization penalty fixed at $\lambda = 0.03$, we can then compare how RMSE posterior means and credible intervals to results without a fixed penalty.

On the training set, the RMSE posterior mean is 41.32 ([9.32, 56.79]), which is a moderate decrease from 47.10 ([19.38, 60.86]) achieved by the model with the non-fixed penalty. However, the 95% credible interval

with the fixed penalty is wider, indicating slightly higher uncertainty. This increased uncertainty agrees with the poor convergence observed in the sampling.

On the testing set, the RMSE posterior mean is 62.60 ([54.97, 75.60]), which is extremely similar to the results of the previous model, which achieved an posterior mean error of 62.59 ([55.19, 73.12]).

Relative to the results of the same model without a fixed regularization penalty, the weakened penalty improves fit to the training data while increasing uncertainty, and it does not significantly impact prediction accuracy. This result demonstrates that adjusting the regularization penalty alone is not enough to recover interpretability of the linear mean within the Gaussian process. Addressing the competition between the Gaussian process and linear component to explain the same patterns in the data may provide a path towards the goal of increased interpretability.

3.3 Orthogonal Gaussian Process with Analytical $\hat{\beta}$

Next, we apply the modified covariance function $k_*(\mathbf{x}, \mathbf{x}')$ introduced by Plumlee and Roshan (2016) to orthogonalize the Gaussian process component with respect to the linear mean. This model seeks to answer the question of whether this restriction the Gaussian process component improves interpretability of linear coefficient estimates.

In this model, we calculate $\hat{\beta}$ using

$$\hat{\beta} = (X^T C_*^{-1} X)^{-1} X^T C_*^{-1} \mathbf{y}$$

We use this form, rather than sampling the coefficients directly, to assess how the orthogonal Gaussian process component interacts with the linear mean. This means that Bayesian Lasso priors are not yet placed on the linear coefficients.

After sampling 6 chains for 10,000 steps, results show that for all sampled parameters $(\ell, \tau, \sigma_{GP}^2, \sigma_n^2)$, $\hat{R} = 1.0$, and both Bulk-ESS and Tail-ESS are significantly greater than 400. Trace plots show stable behavior with good mixing. These diagnostics indicate that the simulation converges successfully, and that the posterior estimates for sampled parameters are good approximations of the target distributions. For each draw, the linear coefficient estimates are calculated and stored. Figure 9 below shows the calculated $\hat{\beta}_0$ value across draws. This figure shows stable behavior, although there are frequent spikes to extreme values. Variation in these values stems from sampled values for σ_n^2 , meaning that uncertainty in $\hat{\beta}$ is driven by the uncertainty in the noise estimate.

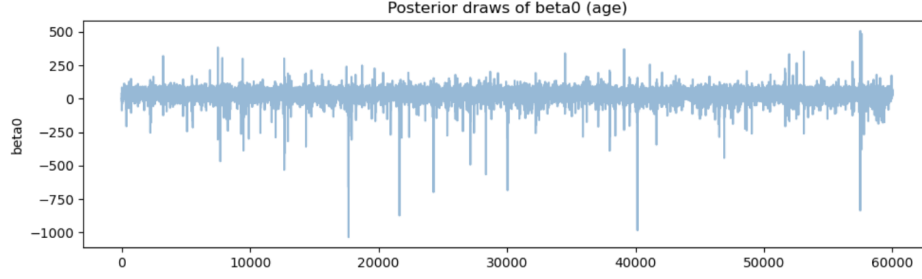


Figure 9: Posterior draws for 'age' (β_0)

Posterior means and 95% credible intervals for $\hat{\beta}$ are calculated, with results shown in Figure 10. In comparison to the posterior estimates of the linear coefficients from the Bayesian Lasso linear model, these estimates show almost identical magnitudes and directions. All coefficients but one correspond to posterior mean estimates with narrow credible intervals that exclude zero. This allows for an intuitive interpretation of the coefficients and the features they relate to. For example, the model shows that the largest positive associations between the set of 10 features and the response are in 's2' and 's5', and the largest negative association is with 's1'. Thus, patients with high 's2' and 's5' measurements at baseline are linked to more severe diabetes outcomes one year later, while the reverse is true for a high 's1' measurement at baseline.

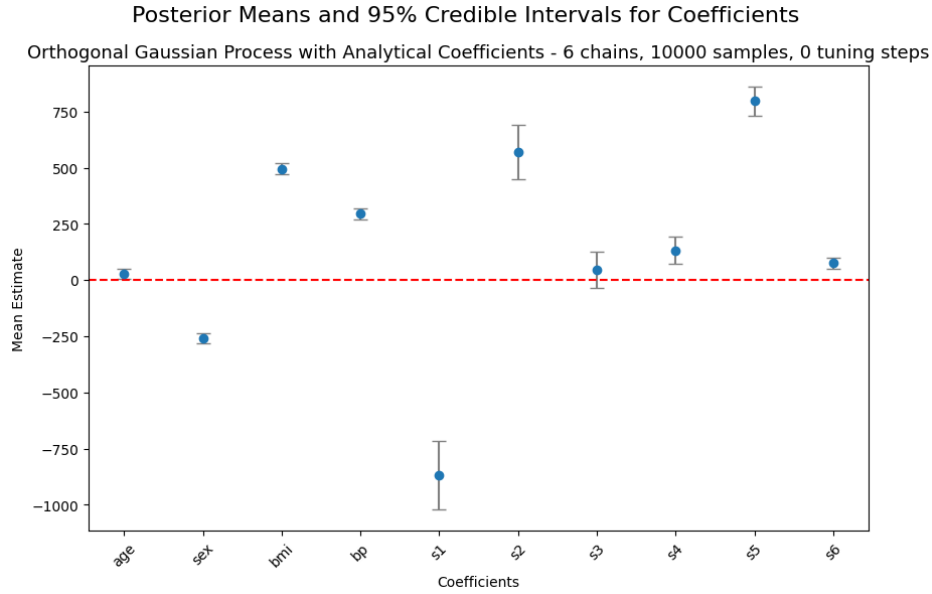


Figure 10: Posterior Means and 95% Credible Intervals for Linear Coefficients

Posterior mean estimates for RMSE are 39.44 ([13.34, 50.04]) for the training set and 57.03 ([53.68, 64.15]) for the testing set. Credible intervals, especially for the testing set, are significantly narrowed and

RMSE estimates are lower compared to the Gaussian process with Bayesian Lasso priors. Compared to the Bayesian Lasso linear model, the training RMSE is lower, but the testing RMSE estimate is similar. The significant drop in training error indicates that the Gaussian process component is fitting to structure in the training residuals, but this structure does not improve generalization to the unseen testing set.

Figure 11 visualizes the Gaussian process and linear mean components of the model separately. Here, as opposed to the Gaussian process with Bayesian Lasso priors, we see that the linear component is contributing to the bulk of the prediction, while the Gaussian process component is contributing on a smaller scale. This meshes with the conceptual understanding of a Gaussian process with a linear mean: the linear mean fits the general linear trend, and the Gaussian process fits non-linear structure within the residuals. In this figure, we see that the Gaussian process contributions increase slightly for larger response values, suggesting that the linear model fits slightly below the trend in the data in this region.

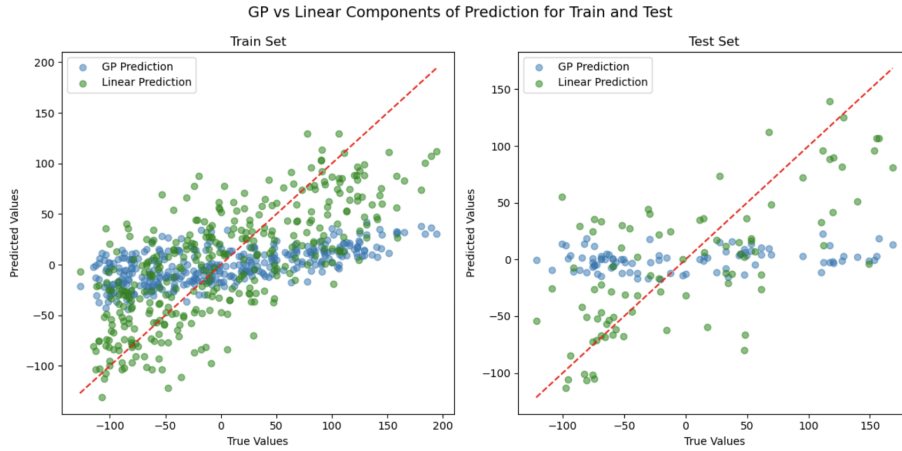


Figure 11: Orthogonal Gaussian Process and Linear Components for Train and Test

These results demonstrate that the orthogonal Gaussian process with a linear mean successfully restores interpretability of the mean function while capturing additional non-linear trends within the data, given that the coefficients are calculated analytically. This model fits more closely to the training data, but does not make any significant impact on prediction error on the unseen testing set.

However, because the linear coefficients are not sampled in this model, the Bayesian Lasso framework is not applied to regularize the coefficients. The question remains: can orthogonality be successfully combined with Bayesian Lasso priors on a linear mean function?

3.4 Orthogonal Gaussian Process with Bayesian Lasso Priors

The final model examined in this study seeks to answer that question by placing Bayesian Lasso priors on the mean of an orthogonal Gaussian process.

For this model, 6 chains with 10,000 steps each are sampled. For the linear coefficients, $\hat{R}$ values are consistent, all within $[1.02, 1.03]$. Bulk-ESS is above 400 for 8 of the coefficients, and above 389 for the remaining 2. Tail-ESS shows slightly worse results, with values for all coefficients falling within $[192, 589]$. Only 4 coefficients show Tail-ESS greater than 400. These diagnostic results are similar to those seen with the non-orthogonal Gaussian process model results, and they indicate small issues with convergence. From trace plots, we observe stationary results, where individual chains will irregularly explore regions independently of the others. Except for the individual chain excursions, the samples show decent mixing. Samples center around zero, indicating heavy-regularization. See Figure 12 for an example.

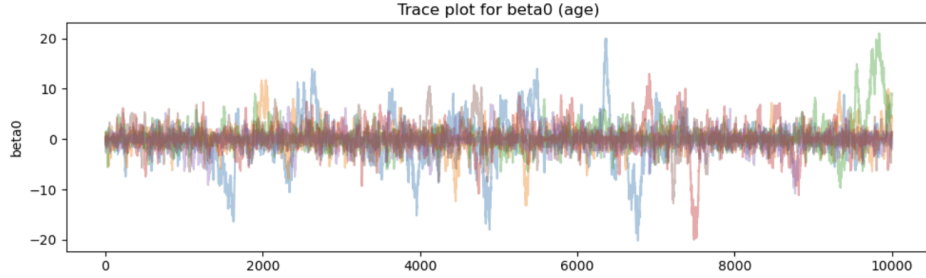


Figure 12: Trace for 'age'(β_0)

The posterior mean estimate of λ^2 is 0.59, with 95% credible interval $[0.001, 1.764]$, meaning that λ is estimated to be 0.77 ($[0.03, 1.33]$). The posterior mean estimate again shows heavy regularization compared to the Bayesian Lasso linear model, and the credible interval shows high uncertainty in the hyperparameter distribution. The heavy regularization is the mechanism that forces linear coefficient posterior mean estimates towards zero, with estimates shown in Figure 13. Additionally, credible intervals are wide and extend symmetrically into both negative and positive regions.

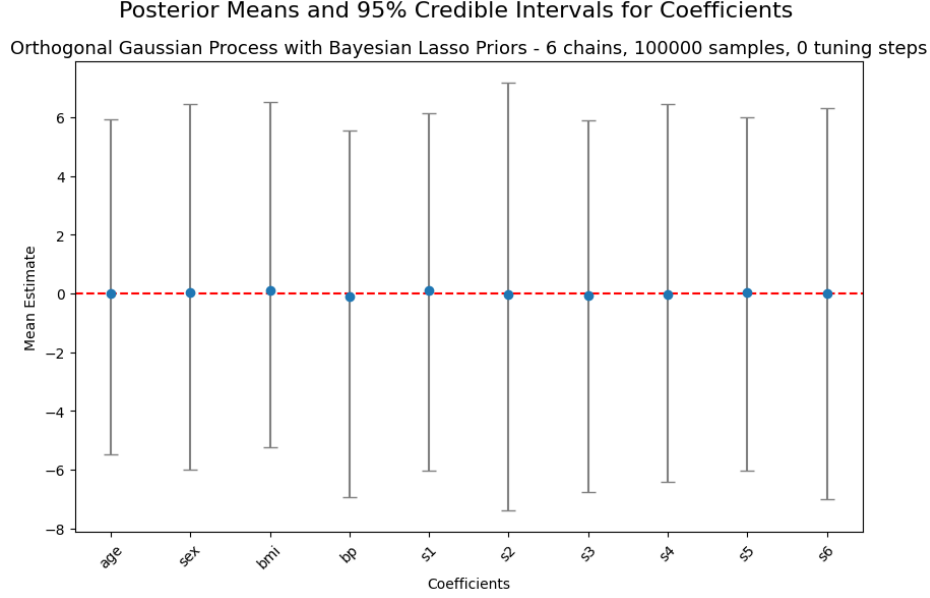


Figure 13: Posterior Means and 95% Credible Intervals for Linear Coefficients

With posterior mean estimates of the prediction error, we compare the model's fit to observed and unobserved data given 10,000 samples and no tuning. On the training set, the RMSE posterior mean estimate is 44.76, with a 95% credible interval represented by [21.61, 57.47]. Compared to the linear model, the RMSE estimate is significantly reduced, although uncertainty is very high. Compared to the non-orthogonal Gaussian process, this estimate is also smaller, although it represents a less significant improvement. The credible interval is also slightly narrower.

On the testing set, the posterior mean estimate of the RMSE is 60.47 ([55.65, 67.44]). This is both larger and contains more uncertainty than the linear model estimate. Similar to the training error, the orthogonal Gaussian process demonstrates slightly lower prediction error and a slightly narrower credible interval compared to the non-orthogonal Gaussian process.

The model once again prefers heavy regularization of the linear coefficients. This results in the linear mean contributing insignificantly to predictions, shown in Figure 14.

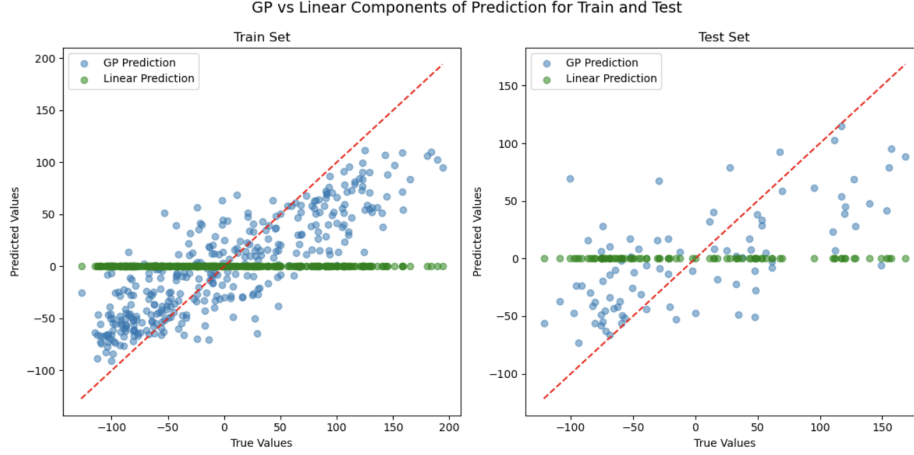


Figure 14: Orthogonal Gaussian Process and Linear Components for Train and Test

These results show that adding in the orthogonality constraint does not lead to improved convergence or interpretability of a Gaussian process with Bayesian Lasso priors, and it only provides a small improvement on generalization to the testing set. However, the linear model’s posterior mean estimate of testing prediction error remains smaller than this, meaning the orthogonal Gaussian process with Bayesian Lasso priors does not offer any advantages on this diabetes dataset, other than a closer fit to the known data.

Fixed Penalty Strength Once more, we test whether fixing the penalty strength to the level of regularization estimate by the Bayesian Lasso linear model ($\lambda = 0.03$) prevents the over-regularization seen in the orthogonal version of the Gaussian process with Bayesian Lasso priors. Previously, we observed that a weak, fixed penalty in the non-orthogonal regularized Gaussian process model only increased uncertainty in both predictions and hyperparameter estimates. However, this case assesses whether the orthogonality restriction will produce different results.

After sampling for 10,000 steps with no tuning period, we can observe the following results. With the regularization penalty fixed at $\lambda = 0.03$, $\hat{R}$ values for the linear coefficient estimates are very large, ranging from $[1.15, 1.77]$, which is a strong indication of poor convergence. In addition, despite 10,000 samples for each chain, meaning 60,000 total samples, the largest Bulk-ESS values for a linear coefficient is 29. Similarly, the maximum Tail-ESS value for a linear coefficient is 37. Trace plots, such as the one shown in Figure 15 below, show very poor mixing, with each chain following an individual path.

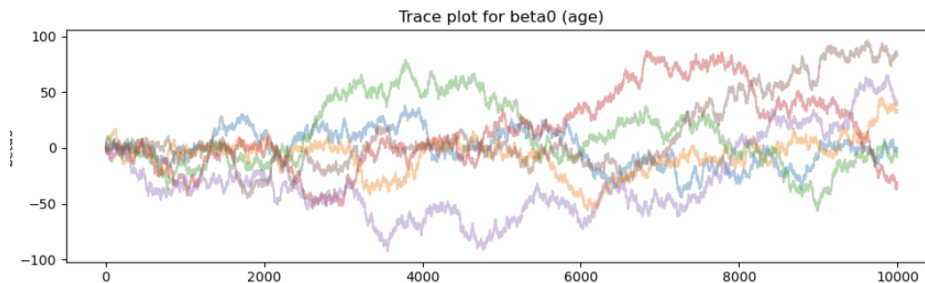


Figure 15: Trace for 'age' (β_0), given $\lambda = 0.03$

Posterior mean estimates of the linear coefficient, shown in Figure 16, show more fluctuation, slightly less symmetry about zero, and significantly more uncertainty than estimates from the model without the fixed penalty. This is to be expected given the poor mixing and wandering of individual chains seen in trace plots. Because all credible intervals contain zero, the model provides no interpretable information about the features.

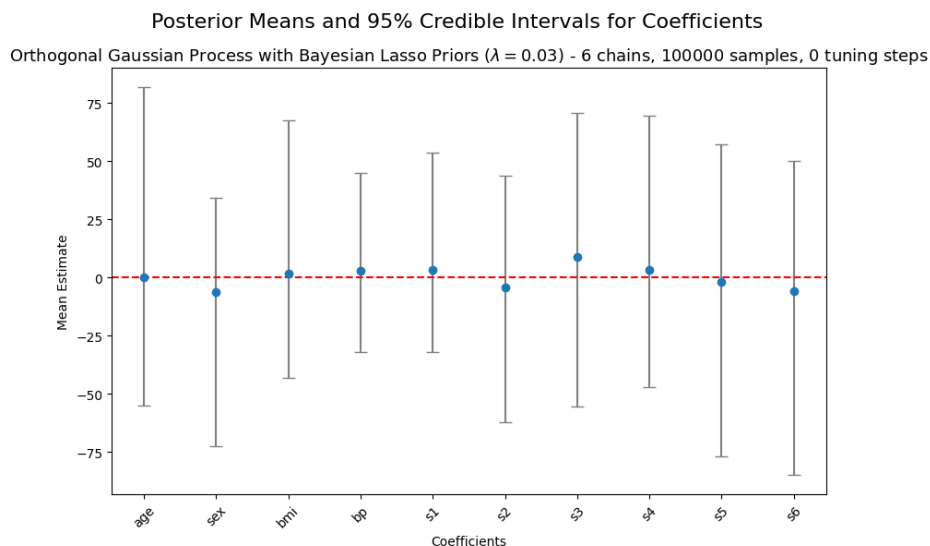


Figure 16: Posterior Means and 95% Credible Intervals for Linear Coefficients

Posterior mean RMSE estimates are then constructed. On the training set, the posterior mean estimate of the RMSE is 41.70 ([15.80, 55.49]). This is a slight improvement over the same model without the fixed penalty, although the uncertainty remains similarly wide. On the unseen testing set, the RMSE posterior mean is 60.49 ([55.55, 68.58]). This is a nearly identical result to the prediction error seen in the same model without a fixed penalty.

These results indicate that when weaker regularization is applied, rather than sampled by the model,

the convergence of linear coefficients deteriorates significantly. Posterior mean estimates of the coefficients are wide and centered around zero. The model fits marginally better to the training data and shows no improvement on predictions on unseen data. This implies that the main difficulty is not over-regularization of the linear mean with the Bayesian Lasso priors, but the difficulty of convergence of the linear mean coefficients, which leaves the Gaussian process component as the only significant contributor to the model’s predictions.

4 Discussion

This study investigates whether Bayesian Lasso priors, alone or in combination with orthogonal Gaussian process methods, can recover interpretability from a linear mean function within Gaussian process regression models. A Bayesian Lasso linear model is used as a baseline, where independent and identical Laplace priors on linear coefficients apply an ℓ_1 -analogous penalty. This model identifies narrow intervals for coefficient estimates and generalizes well to unseen data. A Gaussian process with Bayesian Lasso priors applies heavy regularization that forces the linear coefficients to zero, leaving the Gaussian process component of the model to explain variation in the data alone. When the Gaussian process is orthogonalized with respect to the linear mean function, and coefficients are determined analytically, the model captures linear signal with the mean function and applies the Gaussian process to identify structure in the residuals. However, an orthogonal Gaussian process with Bayesian Lasso priors results in heavy regularization of the linear mean, resulting in the Gaussian process solely contributing to predictions. Even when a weak regularization penalty is fixed, the model is not able to construct meaningful estimates of linear coefficients due to poor convergence.

The results of this paper focus heavily on interpretability, because interpretability is a crucial link between a model and the system it describes. For example, within the diabetes dataset provided by Efron et al. (2004), interpretability provides value because it deepens understanding of the biological mechanisms that are linked to diabetes progression, as opposed to a black-box model that is limited to accurate predictions. While a zero-mean Gaussian process could fit to this data well and make accurate predictions, a linear mean with interpretable coefficients presents information on both the directions and relative magnitudes of associations between blood serum measurements and diabetes outcome after one year.

Moreover, while an accurate and interpretable result may be produced by applying a zero-mean Gaussian process to the residuals of a separate regularized linear regression, the unified approach investigated here presents distinct theoretical advantages over a two-stage approach. The joint Bayesian approach provides uncertainty quantification for all hyperparameters and predictions in a way that properly integrates the uncertainty originating from both models. A two-stage approach could fit a likelihood-based or Bayesian

linear model first, with a Bayesian Gaussian process fit to its residuals. However, in the first case, linear model coefficients are point estimates with no indication of uncertainty. In the second case, the Bayesian linear model would quantify uncertainty for its hyperparameters and its predictions, but the Gaussian process would not be able to incorporate this uncertainty. In this study, the orthogonal Gaussian process with analytical linear coefficients shown in this study presents another path, where the linear coefficients are computed from sampled Gaussian process hyperparameters. However, with this method, the uncertainty in the coefficients originates directly from uncertainty in the sampled hyperparameters. Overall, given that the only model that achieved results comparable to the linear baseline was a two-stage approach that calculated linear mean coefficients from the results of an orthogonal Gaussian process, this study demonstrates that the theoretical advantages of a joint Bayesian approach are overshadowed by the practical performance provided by a two-stage approach.

There are several logical directions that further research into this topic might extend. In this study, prior distributions for hyperparameters were largely chosen based on assumptions about the likely scales of the sampled parameters. A comprehensive evaluation of different distributions was not performed. Comparing results with different distributions of hyperparameters would address this limitation. Secondly, different kernels might be applied within the Gaussian process. Lastly, the results of this paper were shown through the diabetes dataset from Efron et al. (2004), on which a linear model is able to successfully explain a large part of the variation in the data. Testing these models on different datasets may result in the identification of scenarios where a more complex model, such as the orthogonal Gaussian process with Bayesian Lasso priors, may outperform a regularized linear model.

Overall, the most significant result of this study comes from the comprehensive lesson in applying Bayesian analyses. Communicating uncertainty and risk properly is a critical skill across every domain, and the lessons in communication encountered within the process of devising and writing this study will be carried forward with me as I take my next steps academically and professionally.

5 Conclusion

This study investigated methods to combine the flexibility, uncertainty quantification, and predictive strengths of Gaussian process regression with the interpretability of a regularized linear mean, through a detailed comparison of three Gaussian process approaches and a linear baseline on a standardized diabetes dataset. Results showed that Bayesian Lasso priors within the linear mean of a Gaussian process resulted in heavy regularization of the linear coefficients, making the Gaussian process component the only significant driver of prediction. To attempt to prevent the Gaussian process from out-competing the linear mean function

in explaining the signal within the data, an orthogonality constraint was applied with respect to the mean function. An orthogonal Gaussian process model with an analytical and non-regularized linear mean function converged successfully and recovered interpretability of the coefficients. However, with Bayesian Lasso priors on the mean function, linear coefficients were driven to zero. Fixing a weak penalty on the model demonstrated that convergence of the linear coefficients deteriorates without strong regularization.

These results imply that competition between the Gaussian process and linear components may not be the reason the linear mean contributes insignificantly. Rather, it may be the result, where heavy regularization improves the poor convergence of linear coefficients, even as it trivializes the linear component of the model.

References

- Abril-Pla, Oriol et al. (Sept. 2023). “PyMC: A Modern, Comprehensive Probabilistic Programming Framework in Python”. In: *PeerJ Computer Science* 9, e1516. DOI: 10.7717/peerj-cs.1516.
- Ashton, Gregory et al. (Apr. 2019). “Bilby: A User-Friendly Bayesian Inference Library for Gravitational-Wave Astronomy”. In: *The Astrophysical Journal Supplement Series* 241.2, p. 27. DOI: 10.3847/1538-4365/ab06fc. URL: <https://arxiv.org/abs/1811.02042>.
- Efron, Bradley et al. (Apr. 2004). “Least Angle Regression”. In: *The Annals of Statistics* 32.2, pp. 407–499. DOI: 10.1214/009053604000000067. URL: <https://projecteuclid.org/euclid.aos/1083178935>.
- Gelman, Andrew et al. (2014). *Bayesian Data Analysis*. 3rd ed. Boca Raton: CRC Press. ISBN: 9781439898208.
- Park, Trevor and George Casella (June 2008). “The Bayesian Lasso”. In: *Journal of the American Statistical Association* 103.482, pp. 681–686. DOI: 10.1198/016214508000000337.
- Pedregosa, F. et al. (June 2015). “Scikit-learn”. In: *GetMobile: Mobile Computing and Communications* 19.1, pp. 29–33. DOI: 10.1145/2786984.2786995.
- Plumlee, Matthew and Joseph V. Roshan (2016). *Orthogonal Gaussian Process Models*. arXiv: 1611.00203. URL: <https://arxiv.org/abs/1611.00203>.
- Rasmussen, Carl Edward and Christopher K. I. Williams (2006). *Gaussian Processes for Machine Learning*. Cambridge, MA: MIT Press. ISBN: 9780262182539.
- Vehtari, Aki et al. (July 2020). “Rank-Normalization, Folding, and Localization: An Improved $\hat{R}$ for Assessing Convergence of MCMC”. In: *Bayesian Analysis*. DOI: 10.1214/20-BA1221.